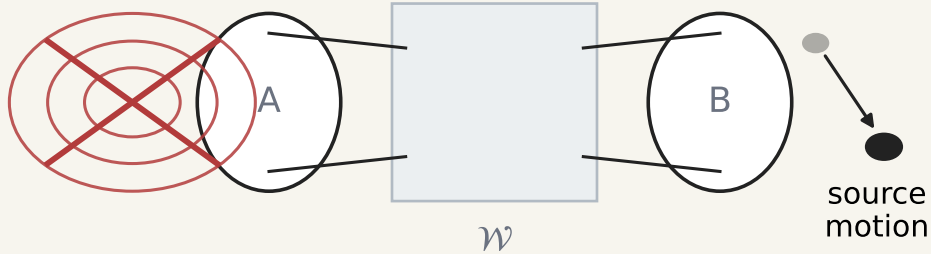


Forbidden channel

source near B moves; no throat flux

monopole fixed no current or stress-energy flux



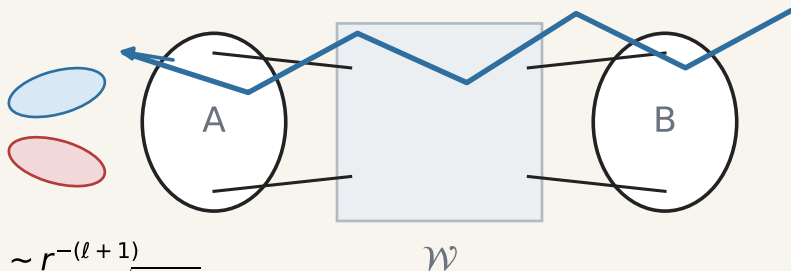
cross-throat $1/r$ induction is absent

Allowed channel

higher multipoles and waves

$$\ell \geq 1$$

waves may scatter



potential $\sim r^{-(\ell+1)}$
finite throat: $\tau_\ell \sim e^{-\sqrt{\ell(\ell+1)}L/R}$

Allowed channel

actual transit or boundary work

$$\Delta E_W \neq 0$$

flux through
worldtube



Brown--York balance
transit or backreaction

geometry $\mathcal{W}$ can do work